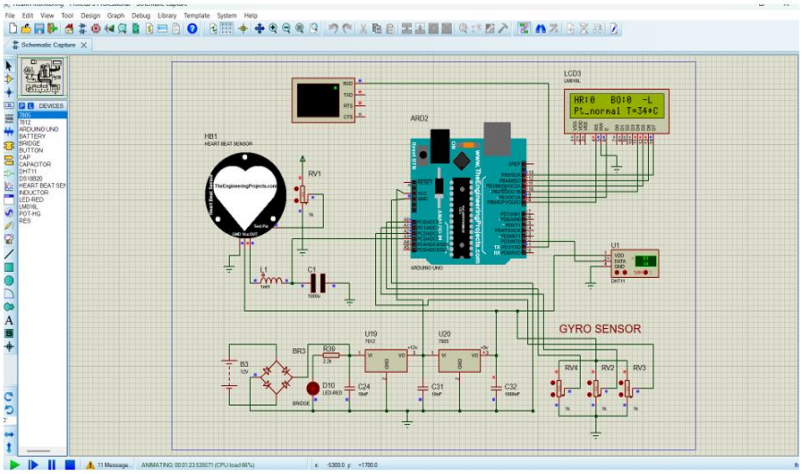
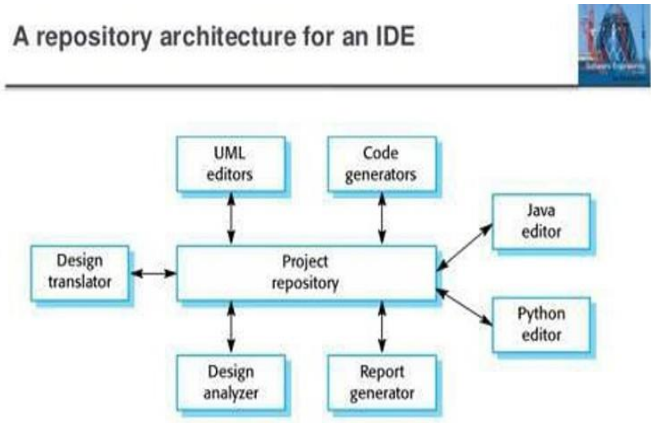
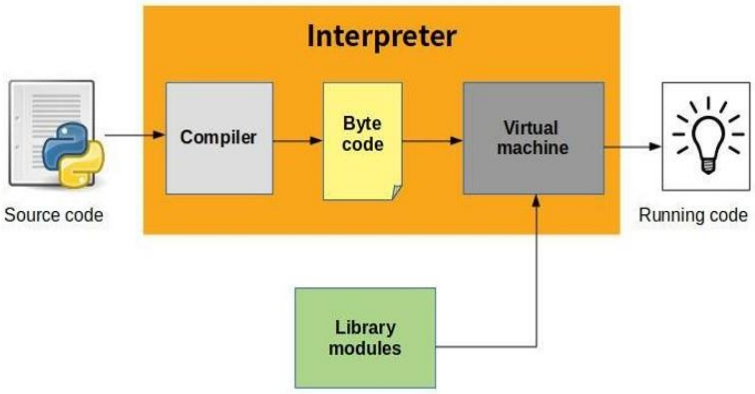
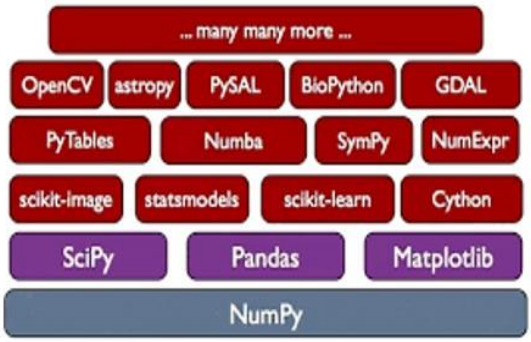
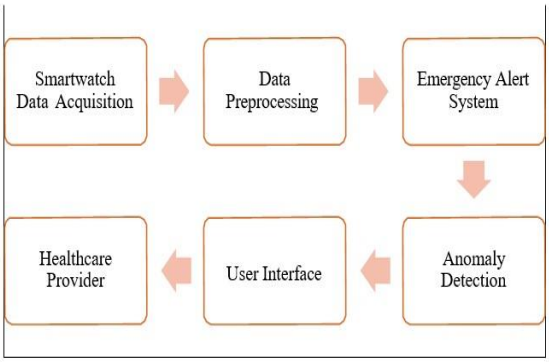
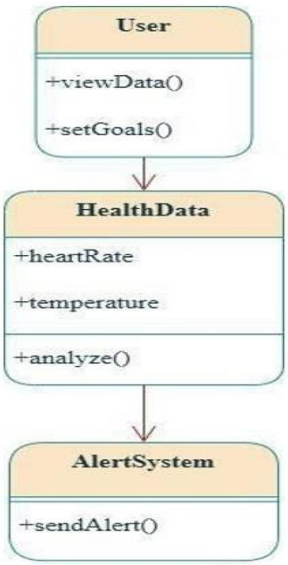

GYRO SENSOR


A repository architecture for an IDE
UML editors
Code generators
Java editor
Design translator
Project repository
Python editor
Design analyzer
Report generator


Interpreter
Source code
Compiler
Byte code
Virtual machine
Running code
Library modules

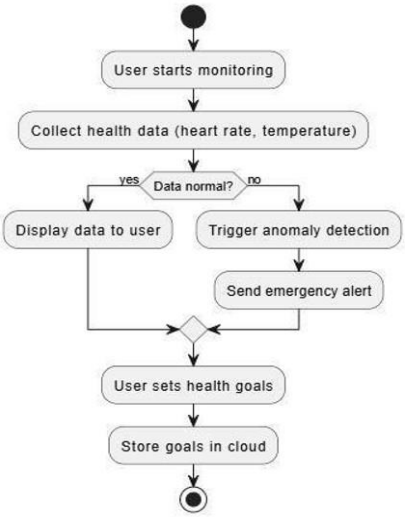

... many many more ...
OpenCV
astropy
PySAL
BioPython
GDAL
PyTables
Numba
SymPy
NumExpr
scikit-image
statsmodels
scikit-learn
Cython
SciPy
Pandas
Matplotlib
NumPy

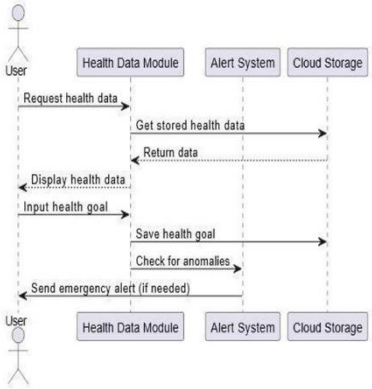

Smartwatch Data Acquisition
Data Preprocessing
Emergency Alert System
Anomaly Detection
User Interface
Healthcare Provider


User
+viewData()
+setGoals()
HealthData
+heartRate
+temperature
+analyze()
AlertSystem
+sendAlert()

View Health Data
Set Health Goals
Track Progress
Receive Alerts
Emergency Alert
User

HTML
CSS


User starts monitoring
Collect health data (heart rate, temperature)
Data normal?
yes
no
Display data to user
Trigger anomaly detection
Send emergency alert
User sets health goals
Store goals in cloud


User
Health Data Module
Alert System
Cloud Storage
Request health data
Get stored health data
Return data
Display health data
Input health goal
Save health goal
Check for anomalies
Send emergency alert (if needed)